

①

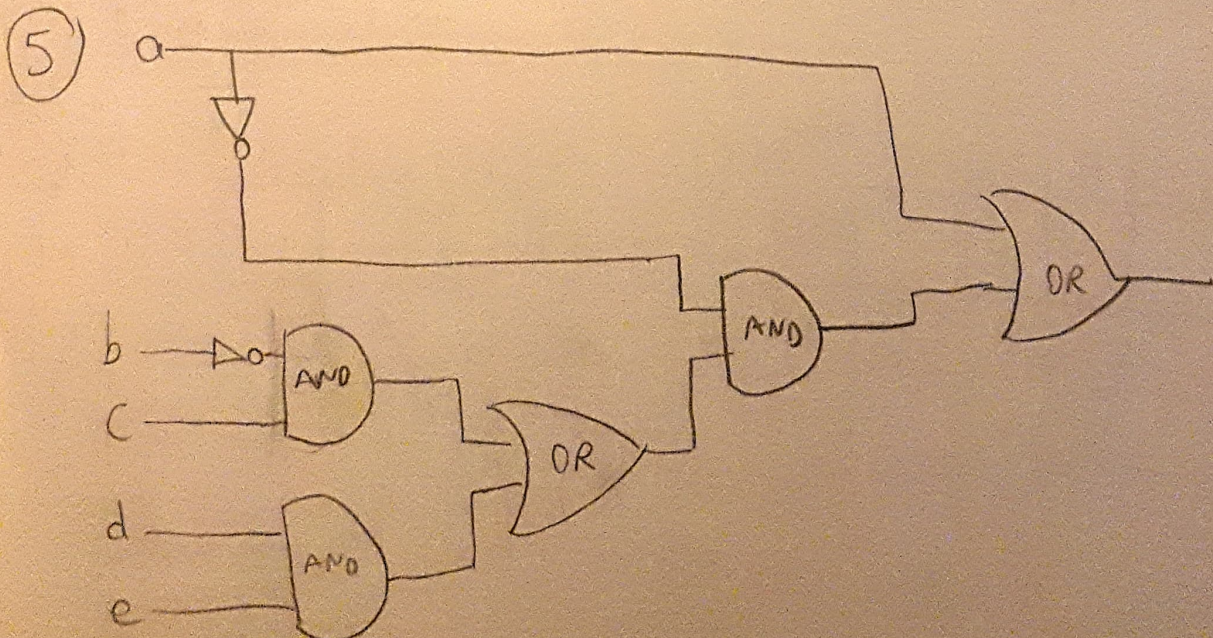
x	y	z	yz	x + yz
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	1	1
1	0	0	0	1
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

x	y	z	x+y	x+z	(x+y)(x+z)
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	1	0	0
0	1	1	1	1	1
1	0	0	1	1	1
1	0	1	1	1	1
1	1	0	1	1	1
1	1	1	1	1	1

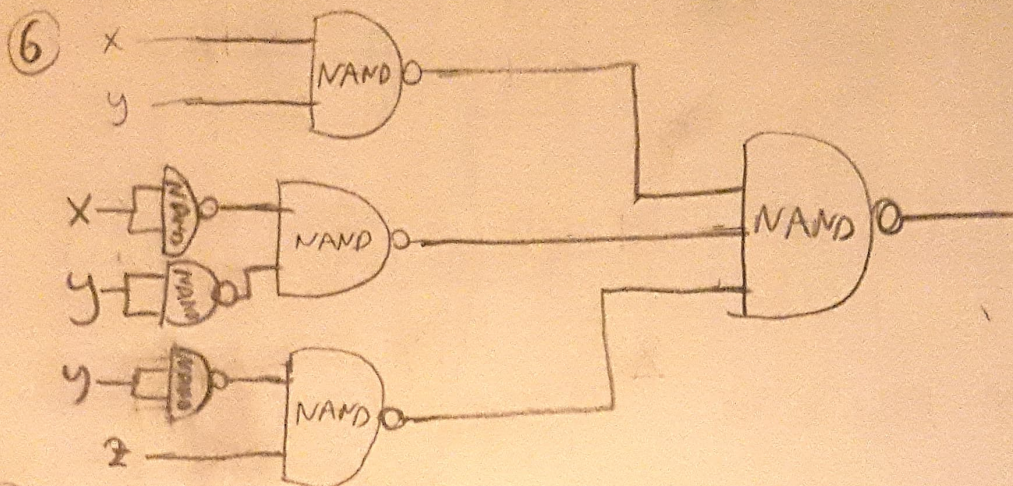
② $a\cancel{a}b' + ac + b\cancel{a}b' + bc + \cancel{c}a'b' + \cancel{c}c = ac + bc + \cancel{c}a'b'$

③ $a'bc + abc + abc' + a'bc' = bc(\underbrace{a' + a}_1) + bc'(\underbrace{a' + a}_1) = bc + bc' = b(\underbrace{c + c'}_1) = \underline{b}$

④ $(a + a'(b'c + de))' = a' (a + ((b + c')(d' + e')))$



Emir Asol
27933 SA



⑦

a	b	c	a'	(a'+b)	(a'+c)	(a'+b)(a'+c)
0	0	0	1	1	1	1
0	0	1	1	1	1	1
0	1	0	1	1	1	1
0	1	1	1	1	1	1
1	0	0	0	0	0	0
1	0	1	0	0	1	0
1	1	0	0	1	0	0
1	1	1	0	1	1	1

$$F = a'b'c' + a'b'c + a'bc' + a'bc + abc \rightarrow \text{minterms}$$

$$F = (a'+b+c)(a'+b+c')(a'+b'+c) \rightarrow \text{maxterms}$$

⑧

x \ z	00	01	11	10
0	0	1	1	0
1	1	1	0	0

$$(y'z + x'z)' = (y+z')(x+z')$$

⑨ a)

x \ z	00	01	11	10
0			1	1
1			1	1

y

b)

AB \ CD	00	01	11	10
00				
01	1		1	1
11			1	
10				

$$A\bar{B}\bar{D} + BCD$$

Emir Asad
27933 &

c) $AB \backslash CD$

	00	01	11	10
00			1	
01			1	
11	1	1	1	
10			1	

$$CD + ABD + ABC$$

d) $wx \backslash yz$

	00	01	11	10
00			1	1
01				
11	1	1	1	1
10				

$$w'x'y + wx$$

10) $AB \backslash CD$

	00	01	11	10
00	1		1	1
01		1	1	
11			1	1
10	1		1	1

$\rightarrow B'D + CD + A'BD + AC$

11) a) $wx \backslash yz$

	00	01	11	10
00	1			1
01	1	1	1	1
11		1	1	
10	1			1

b) $AB \backslash CD$

	00	01	11	10
00	1		1	1
01		1	1	
11			1	1
10	1		1	1

essential pm

1		1	
1		1	

xz'

essential pm

1	1		
1	1		

xz

pm

1		1	
1		1	

$w'z'$

pm

1	1	1	1
1	1	1	1

$w'x$

pm

1		1	
1		1	

essential pm

1	1		
1	1		

pm

	1	1	
	1	1	

essential pm

1	1		
1	1		

pm

	1		
	1		